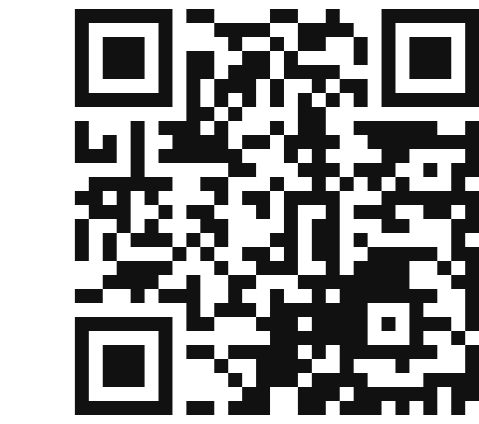


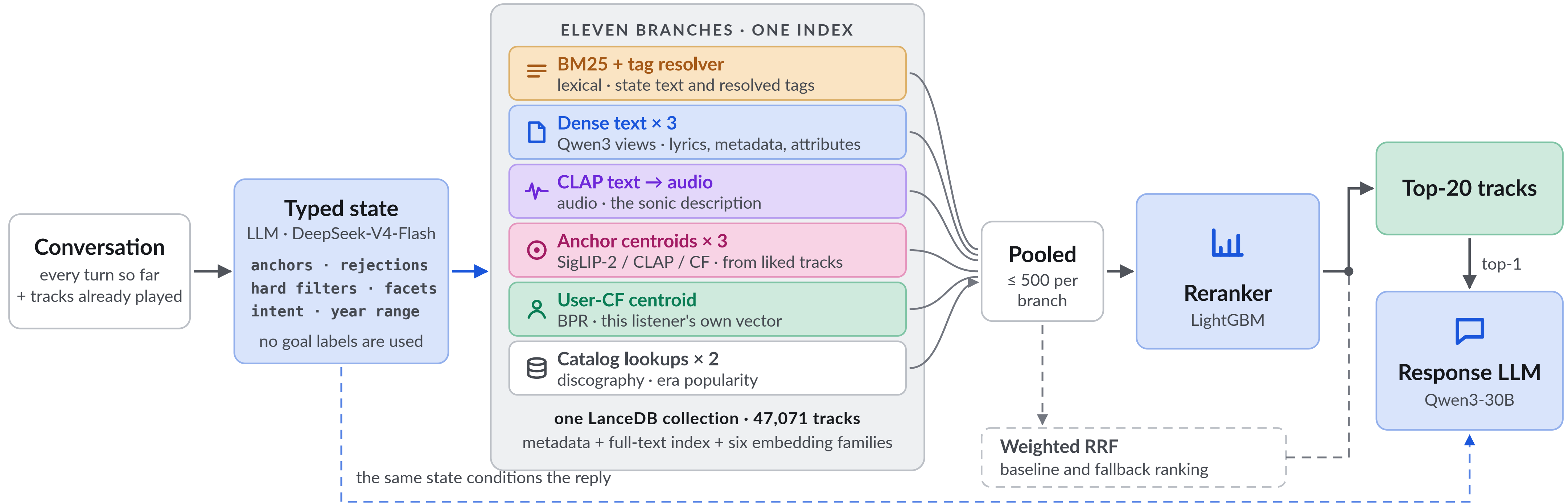
# State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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11 retrieval branches  
146 ranker features  
29/40 where that finished



Paper, poster, code, data  
[npatta01.github.io/music-crs-2026](https://npatta01.github.io/music-crs-2026)



## What the system reads off a conversation

Earlier turns: we played Neko Case’s “Calling Cards” (“*not quite the one*”), then “Man” (“*not it either*”).

**User, now:** “...much more subdued... a specific sense of place in the lyrics... a somewhat stark, almost a cappella delivery.”

EXTRACTED STATE

**anchor artist** Neko Case (must use) · earlier turns  
**rejected** “Calling Cards”, “Man” · earlier turns  
**facets** mood = subdued  
theme = sense of place  
sonic = stark, almost a cappella  
**request** hidden\_target

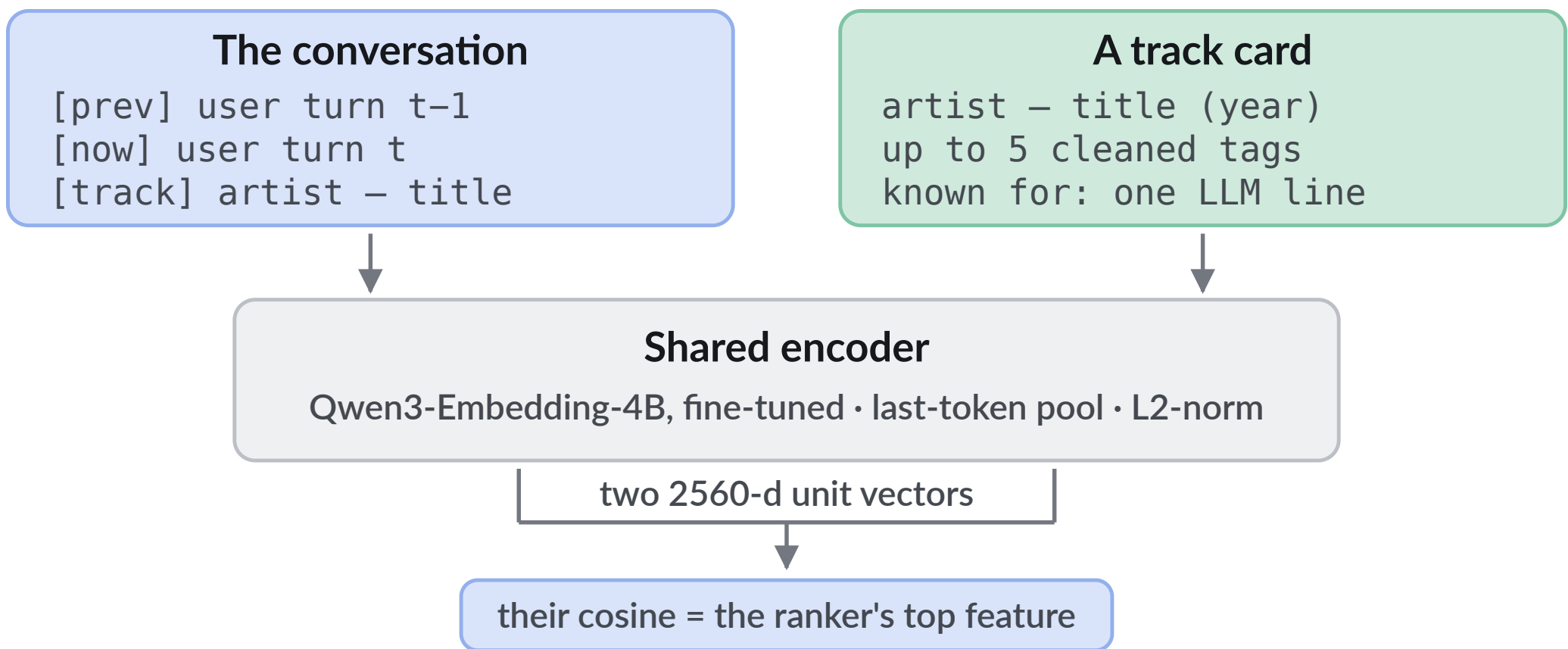
The full schema: request type · intent mode · anchors and pinned tracks · per-track feedback · rejected tracks and artists · hard filters · mood, genre and sonic facets · process constraints · routing flags · lyrical theme · release-year window.

## What the ranker leans on

| FEATURE FAMILY              | SHARE OF GAIN |
|-----------------------------|---------------|
| Bi-encoder cosine           | 59.0%         |
| Per-branch rank and score   | 11.5%         |
| Session and artist affinity | 10.7%         |
| Popularity · state · tags   | 9.2%          |
| Other similarity cosines    | 8.9%          |
| Constraints and rejections  | 0.7%          |

LightGBM LambdaMART · 146 features per (turn, track) pair.

### THE BI-ENCODER BEHIND THAT TOP FEATURE



## How it scored

| OFFICIAL SPLIT              | NDCG@20 | JUDGE | COMPOSITE |
|-----------------------------|---------|-------|-----------|
| Blind-A · development phase | 0.4380  | 4.70  | 0.5799    |
| Blind-B · final phase       | 0.2537  | 3.30  | 0.3811    |

## What is not ideal

We trained and validated the ranker on the same dev turns.

THE LISTENER ASKED FOR

OUR TOP PICK

We extracted more than we used

the Kamelot track  
from Silverthorn

Kamelot — Fallen Star,  
from Haven  
We extracted “Silverthorn” but  
never filtered on it

We used no data from outside the catalog

exactly 126.70 bpm,  
in G minor

the catalog has no tempo or  
key, and we added no  
outside source

The reply said “Got it” for a track we did not have

“Watercolors”  
by Pat Metheny

Metheny — Alfie, with the  
reply “Got it — here’s ‘Alfie!’...”  
“Watercolors” is not in the catalog

## The training labels often ignore the request

**User:** “...similar chill vibe, but *from a different artist?*” (just played: Bonobo)

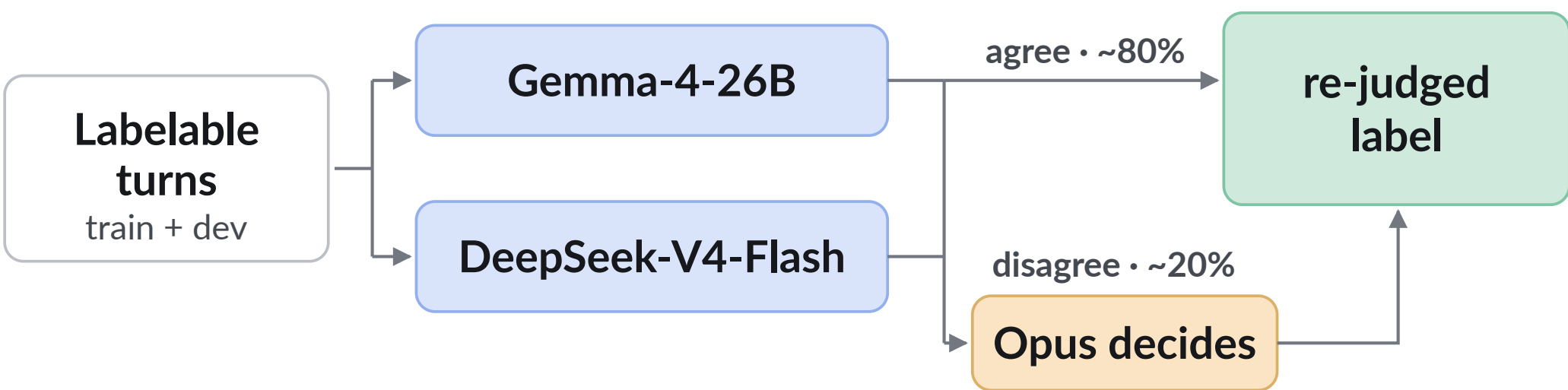
**Target:** Bonobo — “Jets” · annotated MOVES toward goal

**Next turn:** “...hoping for something *from a different artist this time...*”

**Target:** Bonobo — “Pieces” · annotated MOVES toward goal

The label is whatever played next in an LLM-generated session, even when the listener had just asked for something else.

### OUR FIX: TWO LLM JUDGES RE-SCORE THE LABELS



Agreement split measured on the training turns.